PURDUE UNIVERSITY | Institute for Digital and Advanced Agricultural Systems

# IDAAS Newsletter
# September 2025

## Directors' Opening

At this time of year as a new school year starts, I (Ignacio) think on new beginnings and our Institute is on his way of becoming a global hub on Digital Agriculture, but with a major focus on a more important challenge of connecting people and building multidisciplinary teams to transform modern farming. As I reflect on everything that we have achieved as humanity, I feel inspired and excited about the road ahead for our Institute, College of Agriculture, and university. Our affiliates and readers can find our redefined mission and vision statements in our website, https://ag.purdue.edu/idaas/

I (Dennis) love limericks and might make th5is a regular "column" in the IDAAS newsletter. Since we can't go a day without hearing about, likely using AI, here's one for today (of course generated by AI because I am not creative enough to generate these quickly:

*In the office where deadlines are tight,*

*AI lends a hand day and night.*

*It drafts and it plans,*

*Sorts data in spans,*

*Making work feel delightfully light!*

Now with that in your mind, if you are wondering which tool ought to use for some associated tasks you have in mind, maybe check the AI LibGuide by Dr. Jing Lu. She offers a great breakdown of tools by type of work and gives some nice tables showing pros, cons, limitations, pricing, etc.

For sure, don't miss the upcoming events section of this newsletter. That is where you'll find more details about the SyDAG Digital Ag Symposium organized and run by a fabulous group of students with wise counsel from Ignacio Ciampitti and Ankita Raturi.

Wishing you the best,

Dennis R. Buckmaster    Ignacio Ciampitti

## Upcoming Events

### Join Us at SyDAg 2025: Bridging Innovation and Impact in Digital Agriculture

Mark your calendars for October 18–20, 2025, as Purdue University hosts the inaugural Symposium on Digital Agriculture (SyDAg). This student-led event is designed to connect researchers, industry professionals, and students to explore the transformative role of digital technologies in agriculture.

**What to Expect**

- Hackathon (Oct 18–19)**:** Engage in a 48-hour innovation sprint to develop practical solutions for real-world agricultural challenges. Teams will tackle case studies aimed at making digital agriculture more accessible and user-friendly for farmers. SyDAg 2025.
- Symposium (Oct 20)**:** Attend panel discussions, keynote presentations, and poster sessions that delve into the future of digital agriculture. Topics will include scaling innovations, bridging the gap between research and adoption, and fostering cross-sector collaboration.

**Who Should Attend**

SyDAg welcomes individuals from diverse backgrounds, including data science, engineering, agronomy, and environmental science. Whether you're a student, researcher, or industry professional, this event offers valuable insights and networking opportunities.

**Register Now**

Don't miss out on this opportunity to be part of a groundbreaking event in digital agriculture. Register for SyDAg 2025 today and join us in shaping the future of agriculture.

Learn more here!

### IDAAS Fall Seminar Series – Industry Focused

This fall, the IDAAS Seminar Series will feature an exciting lineup of guest speakers from industry who are driving innovation in digital and advanced agriculture. These sessions bring together researchers, students, faculty, and partners to explore cutting-edge ideas, challenges, and opportunities shaping the future of ag technology.

**Upcoming Speakers:**

- Oct. 23 – Eric Schilling, Global Learning and Development Director, *Elanco*
- Nov. 6 – Ryan Bartlett, Executive Innovation Leader & Business Developer, *Pairwise*
- Nov. 20 (Virtual) – Sanjeewa Narayana, IT Technician, *University of Guelph*
- Dec. 4 – Aaron Shew, VP of Product & Data, *Acres*

Seminars will be held at 11:30 AM, in WSLR 116, except for November 20th, which will be held virtually.

More details on their presentations will be shared soon. Learn more about previous seminars and upcoming events at: purdue.ag/IDAAS.

### 2025 September AI Unleashed Series

Our AI Unleashed series kicked off on September 9 with Kenny Wilson, Purdue's Director of Artificial Intelligence and Automation, who shared insights on the university-supported AI tools and services available to faculty, staff, and students. Watch here.

Up next: September 30 | 11:30 AM | WSLR 116

**Dr. Jing Lu, Clinical Assistant Professor in the Purdue Libraries & School of Information Studies – *Exploring AI Responsibly: How to Use the Purdue AI LibGuide***

Artificial Intelligence (AI) is reshaping how we teach, learn, and conduct research. In this session, Dr. Lu will introduce the Purdue AI LibGuide, a curated resource designed to connect students, faculty, and researchers with trusted AI tools, strategies, and examples across disciplines. Attendees will learn how to navigate the guide, explore live AI demonstrations, and gain strategies for evaluating AI's strengths and limitations—while emphasizing responsible and ethical use.

SEPTEMBER 9, 2025
PURDUE IT
Artificial Intelligence
at Purdue University
PURDUE UNIVERSITY

## Graduate Student Spotlight

### Kwabena Bayity - Graduate Peer Coordinator 2025 NSF IoT4Ag AgBridge Summer Program

This summer, Kwabena Bayity, a Ph.D. student in Agricultural Extension Education and Communication, took on a key leadership role as the Graduate Peer Coordinator for the 2025 NSF IoT4Ag AgBridge Summer Cohort Program.

In this position, Kwabena supported interdisciplinary teams of educators and students as they translated advanced agricultural technologies into classroom-ready teaching materials. His guidance, mentorship, and coordination were instrumental in bridging the gap between research and education while fostering collaboration across disciplines.

**Project Highlights – 2025 IoT4Ag AgBridge Summer Cohort**

The AgBridge Program brought together three innovative interdisciplinary projects:

- Spring - a solar-powered remote IoT network gateway for sustainable data collection in agriculture.
- LeafSpec - a portable hyperspectral imaging tool for monitoring crop health and visualizing data.
- micro:bit Cutebot Robotics - hands-on coding, sensors, and automation activities introducing students to precision agriculture concepts.

Each project emphasized hands-on learning, real-world STEM applications, and the integration of agricultural technology into middle school classrooms.

**Kwabena's Perspective**

"Participating in the IoT4Ag AgBridge Summer Cohort Program has been a transformative experience. By helping translate cutting-edge agricultural technologies into classroom-ready teaching materials, I contributed to bridging the gap between research and education. This work supports the development of a future-ready agricultural workforce and promotes STEM engagement among middle school students. As a Ph.D. student in Agricultural Extension Education and Communication, this opportunity has strengthened my skills in interdisciplinary collaboration, curriculum development, and science communication—key areas for my future career in agricultural education and workforce development."

More images from IoT4Ag AgBridge 2025 can be found here.

## Extension Spotlight

### Empowering Indiana Ag: Summer Highlights from Purdue Extension

This summer, Purdue Extension brought hands-on learning and cutting-edge agricultural technology to communities across Indiana. Here are some highlights from the season's impactful events:

**Conservation Equipment Showcase**

Hosted at Clint Orr Farms (Frankfort) and Thompson Farms (Seymour), this showcase featured interactive stations on planting, harvest, spraying, and tillage. With about 40 participants at each site, attendees explored how modern equipment can improve soil health and conservation practices.

**Ag 101: A Foundational Learning Experience**

Held at ACRE, NEPAC, and SEPAC, these sessions welcomed about 30 participants per location. The program offered crop growth staging, pest identification, a month-by-month overview of farming activities, and insights into planting and harvesting equipment—helping participants better understand the agricultural calendar.

**DPAC Field Day: Maximizing Efficiency**

Drawing around 200 attendees, this field day focused on equipment optimization for tighter margins. Topics included combine settings, residue management, thresher adjustments, and weed seed destruction attachments—highlighting how precision and innovation can drive profitability and sustainability.

**Hancock County UAV Part 107 Training**

Nine participants gained hands-on experience with FAA Part 107 regulations, drone operation best practices, and a flight lab that included a spray drone demo—preparing them for safe, effective UAV use in agriculture.

**UAVs at the Indiana State Fair**

On August 17, UAVs took the spotlight at the Purdue Extension Building, where hundreds of fairgoers engaged with experts on sensor and applicator drones, policies, and regulations. The exhibit highlighted the growing role of UAVs in farming.

**Digital Ag at the Indiana Ag Policy Summit**

At the July 30 summit in Boone County, more than 200 attendees explored the impact of digital agriculture on shaping policy and practice for the future of farming.

Make your data work for you
TAKE THE NEXT STEP
PURDUE UNIVERSITY
purdue.ag/digitalag
Purdue On The Farm

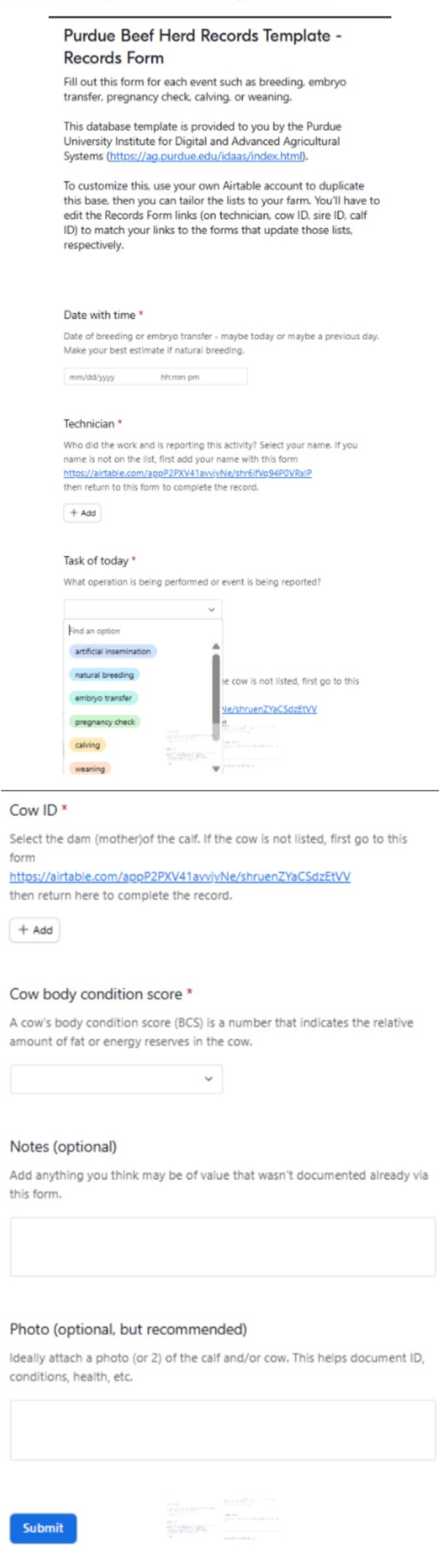
### New Template Available: Digital Beef Herd Records

We are pleased to announce a new addition to the Private Databases template set. This template is designed for digital beef herd records, including breeding, pregnancy checks, calving, weaning, treatments, and more.

Built with extensive data validation and conditional requests, the template streamlines record keeping and ensures information stays relevant to the task at hand.

You can access the template and all the resources needed to duplicate and customize it here: Digital Beef Herd Records. A set of 35 minutes of video tutorials is also available, providing step-by-step guidance to help you digitize and manage these records with confidence.

For other templates and the tutorial on building a private database for a wide range of applications, visit: Private Databases for Agriculturalists.



Purdue Beef Herd Records Template - Records Form

Fill out this form for each event such as breeding, embryo transfer, pregnancy check, calving, or weaning.

This database template is provided to you by the Purdue University Institute for Digital and Advanced Agricultural Systems (https://ag.purdue.edu/idaas/index.html).

To customize this, use your own Airtable account to duplicate this base, then you can tailor the lists to your farm. You'll have to edit the Records Form links (on technician, cow ID, sire ID, calf ID) to match your links to the forms that update those lists, respectively.

Date with time *

Date of breeding or embryo transfer - maybe today or maybe a previous day. Make your best estimate if natural breeding.

mm/dd/yyyy hh:mm pm

Technician *

Who did the work and is reporting this activity? Select your name. If you name is not on the list, first add your name with this form https://airtable.com/appP2PXV41avvjyNe/shr6ifVo94P0VRxIP then return to this form to complete the record.

Task of today *

What operation is being performed or event is being reported?

Cow ID *

Select the dam (mother)of the calf. If the cow is not listed, first go to this form https://airtable.com/appP2PXV41avvjyNe/shruenZYaCSdzEtVV then return here to complete the record.

Cow body condition score *

A cow's body condition score (BCS) is a number that indicates the relative amount of fat or energy reserves in the cow.

Notes (optional)

Add anything you think may be of value that wasn't documented already via this form.

Photo (optional, but recommended)

Ideally attach a photo (or 2) of the calf and/or cow. This helps document ID, conditions, health, etc.

## Research Spotlight

### Purdue Team Awarded Inaugural AI Fusion Grant to Advance Beef-on-Dairy Research

An interdisciplinary team at Purdue University has received the first-ever AI Fusion grant to develop new AI-driven tools for evaluating carcass quality in beef-on-dairy cattle. The project is led by Drs. Hinayah Rojas (Animal Sciences), Qi Guo (Elmore Family School of Electrical and Computer Engineering), and Joseph Campbell (Department of Computer Science), with support from Drs. Jian Jin (Agricultural and Biological Engineering), Brad Kim (Animal Sciences), and Aaron Ault (Electrical and Computer Engineering).

The team will build on Purdue's LeafSpec hyperspectral technology, integrating genomics, imaging, and machine learning to create rapid, scalable methods for assessing meat quality traits. This innovation comes at a critical time: beef-on-dairy crossbreeding has grown dramatically—from 50,000 calves in 2014 to more than 3 million in 2024—creating an urgent need for tools to guide breeding and management decisions.

By combining expertise in animal science, computer vision, machine learning, and robotics, the Purdue team aims to accelerate genetic progress in livestock systems and provide producers with cutting-edge resources to improve efficiency and profitability.

### Faster Corn Yield Predictions with Field Images

**By: Pedro Cisdeli, Gustavo Santiago, German Mandrini, and Ignacio Ciampitti** - Department of Agronomy and IDAAS, Purdue University

Estimating corn yields before harvest helps farmers make better decisions on inputs and harvest planning. Traditionally, this requires walking fields and counting kernels—time-consuming and labor-intensive.

Our team tested a new approach: using smartphone and depth camera images of ears in the field to measure ear traits and forecast yields. A computer vision model isolates the ear from the background, even under tough field conditions. Results showed strong agreement between predicted and harvested yields.
This innovation opens the door to smartphone apps and automated scouting tools, giving farmers faster, earlier insights into yield potential.

**Learn more here:** Ciampitti Lab – Maize Ear Sensing Project

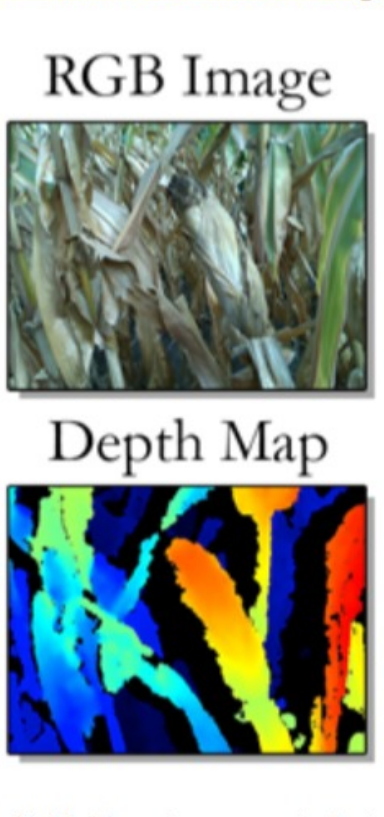


**Figure 1.** Example of a maize ear in the field. Top shows a photo taken with a regular camera. Bottom shows the corresponding depth map. Depth information is what allows us to calculate the ear's size directly in the field.

### IDAAS Publication Database is now Available

**Did you know?** IDAAS has a publication database. You can find it on our publications page. We encourage you to sort, filter, search to find Purdue IDAAS work of interest. As importantly, if you are an IDAAS affiliate, we would like to add your publications to this database. Simply fill out this form that includes the year, full citation, and applicable topic areas. We look forward to showcasing your recent publications in a special section of upcoming newsletters.

We are also adding a brief (3-minute) synopsis of key articles in this database. We are using Google's Notebook LM with a specific prompt to generate a script; then the synopsis is provided in the actual voice of the researcher. You will see these show up in the database. If you want us to generate a script for you to read and record, reach out to digitalag@purdue.edu with the topic of "research synopsis" and include a link to the publication you would like to highlight.

## Industry Engagement

### AGCO Global Agronomy Team Visits Purdue Agriculture and IDAAS Faculty

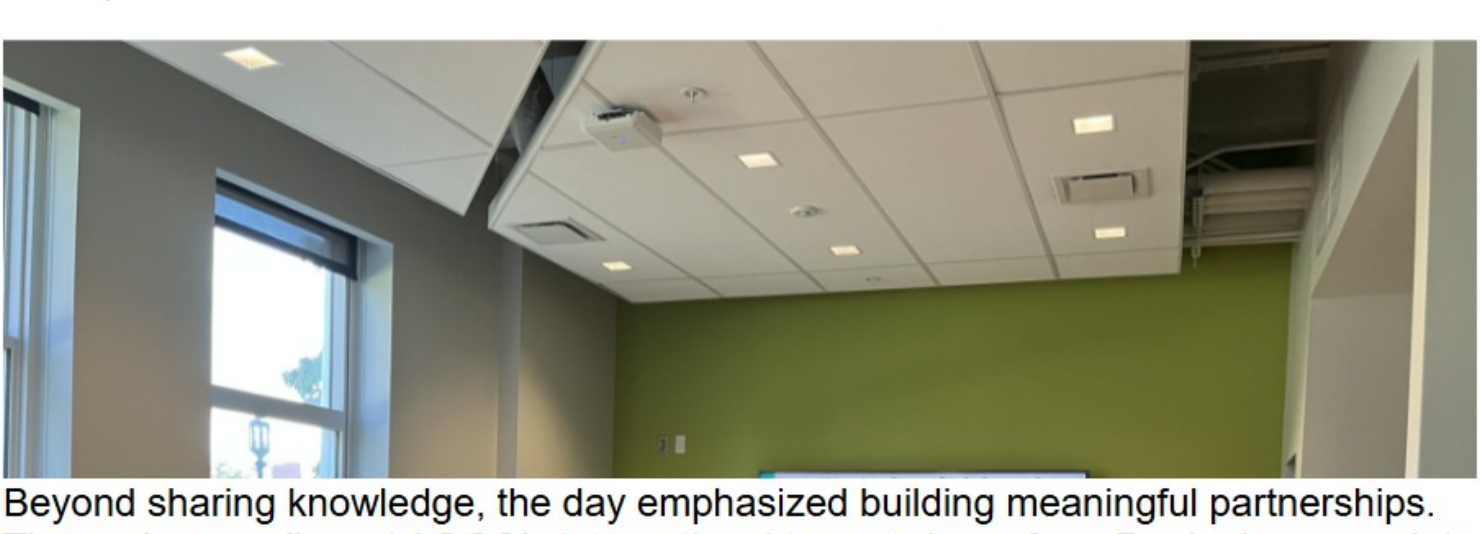
On August 26, Purdue Agriculture and IDAAS welcomed the AGCO Global Agronomy team for a full day of collaboration, discussion, and relationship-building. AGCO, headquartered in Duluth, Georgia, is a worldwide leader in the design, manufacture, and distribution of agricultural machinery and precision ag technology, with operations in more than 35 countries. Their visit to Purdue highlighted the importance of connecting global industry leaders with academic expertise to address the rapidly evolving challenges facing agriculture.

The delegation included Darren Goebel, Nils Zehner (Switzerland), Jens Christian Skov Jensen (Denmark), Jakob Hicks (Minnesota), and Eduardo Sousa (Brazil). Together with Purdue faculty, they explored issues central to farmer profitability, current and future agronomic challenges, and opportunities for innovation. Discussions featured faculty experts such as James Krogmeier and Andrew Balmos (ECE), Mohit Verma (ABE), Dennis Buckmaster (ABE), Bryan Young (Botany & Plant Pathology), Betsy Bower (Agronomy), Ignacio Ciampitti (Agronomy), and Joana Colussi (Agricultural Economics).

Beyond sharing knowledge, the day emphasized building meaningful partnerships. The exchange allowed AGCO's international team to learn from Purdue's research in digital agriculture, weed science, crop management, and agricultural economics, while faculty gained insights into the global perspectives and priorities of a leading agribusiness.

The afternoon included a visit to Purdue's ACRE facility and the Diagnostic Training Center, where Andrew Westfall led hands-on discussions on agronomic diagnostics and field training. The visit concluded with a dinner at McGraw's Steakhouse, capping off a day focused on strengthening ties between Purdue and AGCO.

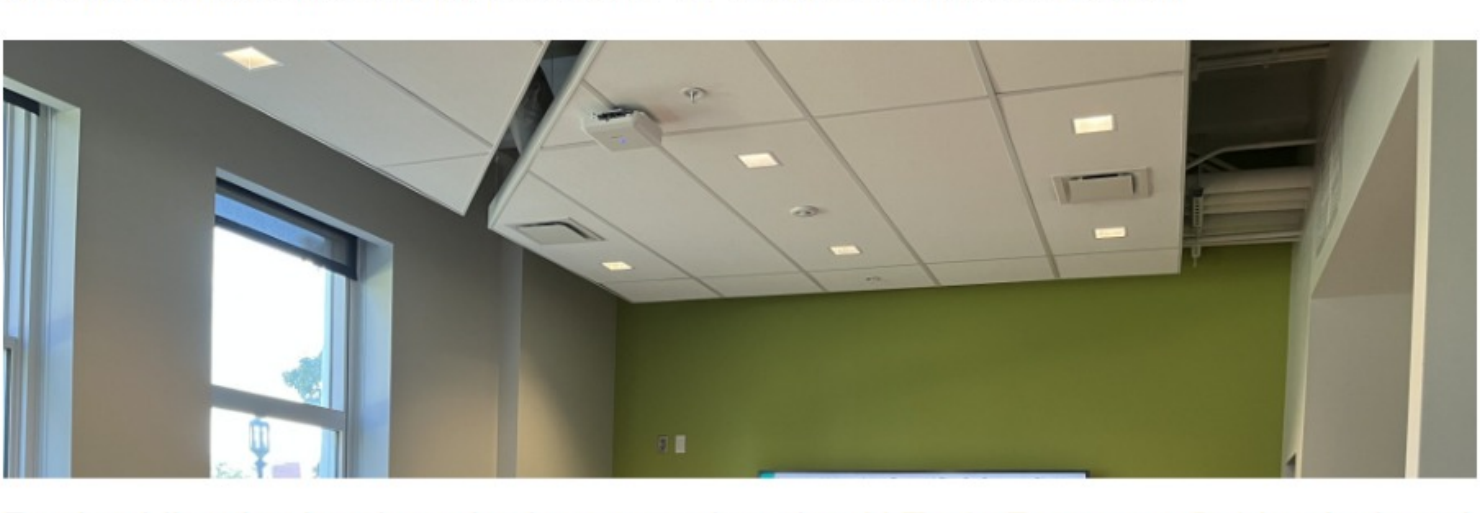
This visit reflects the shared commitment of both AGCO and Purdue to advancing agricultural innovation, learning from one another, and working together to support a more profitable and sustainable future for farmers around the world.



Beyond sharing knowledge, the day emphasized building meaningful partnerships. The exchange allowed AGCO's international team to learn from Purdue's research in digital agriculture, weed science, crop management, and agricultural economics, while faculty gained insights into the global perspectives and priorities of a leading agribusiness.

The afternoon included a visit to Purdue's ACRE facility and the Diagnostic Training Center, where Andrew Westfall led hands-on discussions on agronomic diagnostics and field training. The visit concluded with a dinner at McGraw's Steakhouse, capping off a day focused on strengthening ties between Purdue and AGCO.

This visit reflects the shared commitment of both AGCO and Purdue to advancing agricultural innovation, learning from one another, and working together to support a more profitable and sustainable future for farmers around the world.



Purdue Libraries has launched a comprehensive AI Tools Resource Guide, designed to support researchers, educators, and students in navigating the evolving landscape of artificial intelligence.

**What You'll Find:**

- AI Tools for Literature Review: Discover how AI can streamline the literature review process, enhancing efficiency and depth.
- AI Tools for Writing: Access tools that assist in drafting, editing, and refining written content.
- AI Tools for Systematic Review: Learn about AI applications that aid in conducting systematic reviews with greater precision.
- AI Tools for Teaching and Learning: Explore resources that integrate AI into educational practices, fostering innovative teaching methods.
- How to Cite AI-Generated Content: Understand the best practices for citing AI-generated materials in your work.

Whether you're embarking on a new research project, enhancing your teaching strategies, or seeking to understand AI's role in academia, this guide offers valuable tools and insights.

Don't forget to join us for the upcoming session on September 30 with Jing Lu from Purdue Libraries, where she'll showcase these AI tools in action.

## Connect with IDAAS as an Affiliate

The Institute for Digital and Advanced Agricultural Systems (IDAAS) invites Purdue faculty and staff to become affiliates and join our growing network of researchers and innovators. Affiliates are engaged in areas such as digital agriculture, AI, advanced sensing, data analytics, automation, and related technologies that are shaping the future of agriculture and food systems.

The challenges in these fields are complex and rapidly evolving. Tackling them requires a collaborative, transdisciplinary approach—one that aligns closely with Purdue's Land Grant Mission of building human capital, advancing solutions to pressing global challenges, and partnering deeply with industry and community stakeholders.

To be included in the IDAAS directory of faculty and staff affiliates, please complete the short survey below:

- New affiliates: fill out the form to be added.
- Current affiliates: use the same form to update your information.

Affiliate Form – IDAAS Directory

## Positions Available

Purdue Agricultural and Biological Engineering is looking for a clinical assistant professor of machine systems. This position is teaching and extension with focus on field machinery and digital technologies. See the position description and apply here.

**Digital Identity**

What is Digital Agriculture? It's the use of digital devices to gather, process and analyze spatial and/or temporal data. This data can then guide targeted actions to improve agricultural efficiency, productivity and sustainability.

Contributions to this newsletter are welcomed. Please send your ideas, ideally about one paragraph with links and a photo, to digitalag@purdue.edu.